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Bio-assisted integration of a Co–N/S coordination network with functionalized graphene oxide for supercapacitor and fluorescence sensing applications

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ABSTRACT

A sustainable and eco-friendly strategy was developed to construct a bio-functionalized cobalt–nitrogen/sulfur coordinated graphene hybrid (Bio@Co–N/S@HGO) through a latex-mediated coordination–functionalization approach. In this system, Co^{2+} ions coordinate with thiosemicarbazide through N and S donor atoms, while graphene oxide serves as a high-surface-area support for the resulting coordination component. Biomolecules from *Calotropis gigantea* latex are incorporated during the functionalization process, providing additional surface functionalities and interfacial interactions. Structural and compositional analyses by PXRD, FESEM, XPS, Raman spectroscopy, and FT-IR confirmed the formation of the hybrid architecture. The material exhibited pronounced pseudocapacitive behavior, delivering a specific capacitance of 598.4 F g^{-1} at 10 mV s^{-1} with ~94% capacitance retention after 5000 charge–discharge cycles. This electrochemical response is attributed to the integrated effects of the redox-active Co–N/S component, functionalized graphene framework, and improved accessibility of electroactive sites. The hybrid also exhibited a fluorescence turn-on response toward Al^{3+} under the investigated conditions. This response is interpreted as an Al-associated interaction with the functional sites of the hybrid, while contributions from Al hydrolysis and aggregation cannot be completely excluded. As a latex-free analogue was not investigated, the specific contribution of latex to the electrochemical properties cannot be independently quantified; therefore, the observed performance reflects the combined characteristics of the complete bio-functionalized Co–N/S@graphene hybrid.

1. Introduction

The increasing global energy demands, a rapid depletion of fossil fuel resources, and growing ecological issues have accelerated the development of advanced materials for efficient energy storage and conversion. In this context, the design of high-performance, cost-effective, and sustainable electrode materials has become a central focus. [1,2] Among the diverse energy storage technologies, supercapacitors have emerged as promising candidate due to their high power density, rapid charge–discharge capability, and excellent cycling stability, which makes them highly suitable for portable electronics, hybrid vehicles, and backup systems [3].Supercapacitors, also termed as electrochemical capacitors, serves as intermediate energy storage devices between conventional capacitors and batteries by combining rapid energy delivery with long operational lifetimes [4–6]. Charge storage in supercapacitors occurs via pseudocapacitance and electrical double-layer capacitance (EDLC). EDLC involves electrostatic accumulation of charge at the interface of electrode and electrolyte without faradic reactions, ensuring superior durability. Consequently, carbon-based materials are widely employed due to their high conductivity, large specific surface area, and

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chemical stability [7–9].

Among these, graphene and its derivatives, especially graphene oxide (GO), have gained considerable interest owing to their unique two-dimensional structure and tunable physicochemical characteristics. [10,11] The presence of oxygenated functionalities, including hydroxyl (-OH), carbonyl (-C=O), and epoxy moieties, makes GO a versatile platform for further functionalization and hybrid material design. Graphene-based materials have demonstrated wide applicability in optoelectronics, energy storage, sensing, and nanocomposites. Controlled functionalization of GO enables precise tuning of its electronic structure, surface chemistry, and interlayer spacing, thereby enhancing ion transport and electrochemical performance [12–14].Both covalent and non-covalent functionalization strategies have been investigated; however, achieving an optimal balance between conductivity, stability, and electrochemical activity remains challenging.

Recent studies further show that heteroatom configuration and carbon architecture jointly influence electrochemical behavior. N/O co-doping can enhance surface polarity and electrolyte affinity, whereas BCN-containing hierarchical structures and N-doped porous carbons provide tunable electronic environments and accessible pathways for ion transport and charge storage. In the present Bio@Co–N/S@HGO system, N/S functionalities introduced through bio-assisted modification and Co–thiosemicarbazide coordination provide polar interaction sites, while the graphene framework supports electron transport and suppresses sheet restacking [15–20]. The combination of N/S functionalities with redox-active Co centres therefore provides a distinct approach to coupling surface chemistry, conductivity, and pseudocapacitive charge storage. [21]

Previous studies have established that controlled functionalization of graphene using bio-derived and heteroatom-rich modifiers can effectively tailor its interfacial properties. For example, Yang et al. reported that tannic acid hybridization enhances structural integrity through covalent, hydrogen-bonding, and π–π interactions, while Xu et al. demonstrated that urushiol functionalization improves dispersion and integration with polymer matrices. Similarly, Zhang et al. showed that nitrogen-rich graphene oxide prepared through nucleophilic substitution and amidation exhibits enhanced adsorption performance. Recent studies on graphene-based hybrid architectures have demonstrated that conductive carbon frameworks and interfacial coupling can facilitate charge transport, ion diffusion, and electrochemical charge storage. [22–24]In this context, the present work introduces a distinct latex-mediated strategy in which *Calotropis gigantea*-functionalized graphene oxide is integrated with a Co–N/S coordination framework.The graphene network provides conductive pathways, while the bio-derived functionalities facilitate interfacial interactions with the coordination framework, offering a sustainable approach to constructing electroactive graphene-based hybrids.

In this regard, plant-derived reducing and functionalizing agents have emerged as sustainable alternatives for tailoring GO-based hybrid materials. Latex obtained from *Calotropisgigantea*, a readily available medicinal plant, is of particular interest due to its rich composition of bioactive compounds, including proteins, alkaloids, flavonoids, and terpenoids. [25]These constituents can effectively modulate the surface chemistry of GO by introducing functional groups, increasing interlayer spacing, and improving dispersibility and chemical stability in aqueous media. Moreover, the biodegradable and renewable nature of *C. gigantea* latex aligns with the principles of green chemistry, offering an environmentally friendly route for material synthesis. Its complex mixture of bioactive molecules provides multifunctional interaction sites that can enhance interfacial properties and electrochemical performance, highlighting its potential in advanced energy materials [26,27].

Although our previous studies reported functionalized GO for fluorescence-based sensing of metal ions such as Ti^{4+}, those systems primarily relied on surface defect engineering and organic functional groups to tune optical properties. [28] In contrast to our previous studies on functionalized graphene-based materials for energy storage and fluorescence sensing, the present work explores a bio-assisted route for integrating latex-functionalized graphene oxide with a cobalt–thiosemicarbazide coordination framework. The novelty lies in combining bio-derived surface functionalities, Co–N/S coordination sites, redox-active cobalt centers, and a graphene framework within a single hybrid architecture. *Calotropis gigantea* latex is employed as a bio-derived functionalization component, while the Co–N/S framework provides additional metal- and heteroatom-containing active sites. The resulting hybrid is investigated for electrochemical energy storage and Al^{3+} fluorescence sensing. Importantly, because a latex-free control and independent characterization of the latex-derived organic fraction were not performed, its individual contribution to the electrochemical properties cannot be quantitatively separated. Therefore, the observed performance is attributed to the combined characteristics of the graphene framework, Co–N/S coordination component, and bio-derived functionalization.

2. Experimental section

2.1. Materials and methods

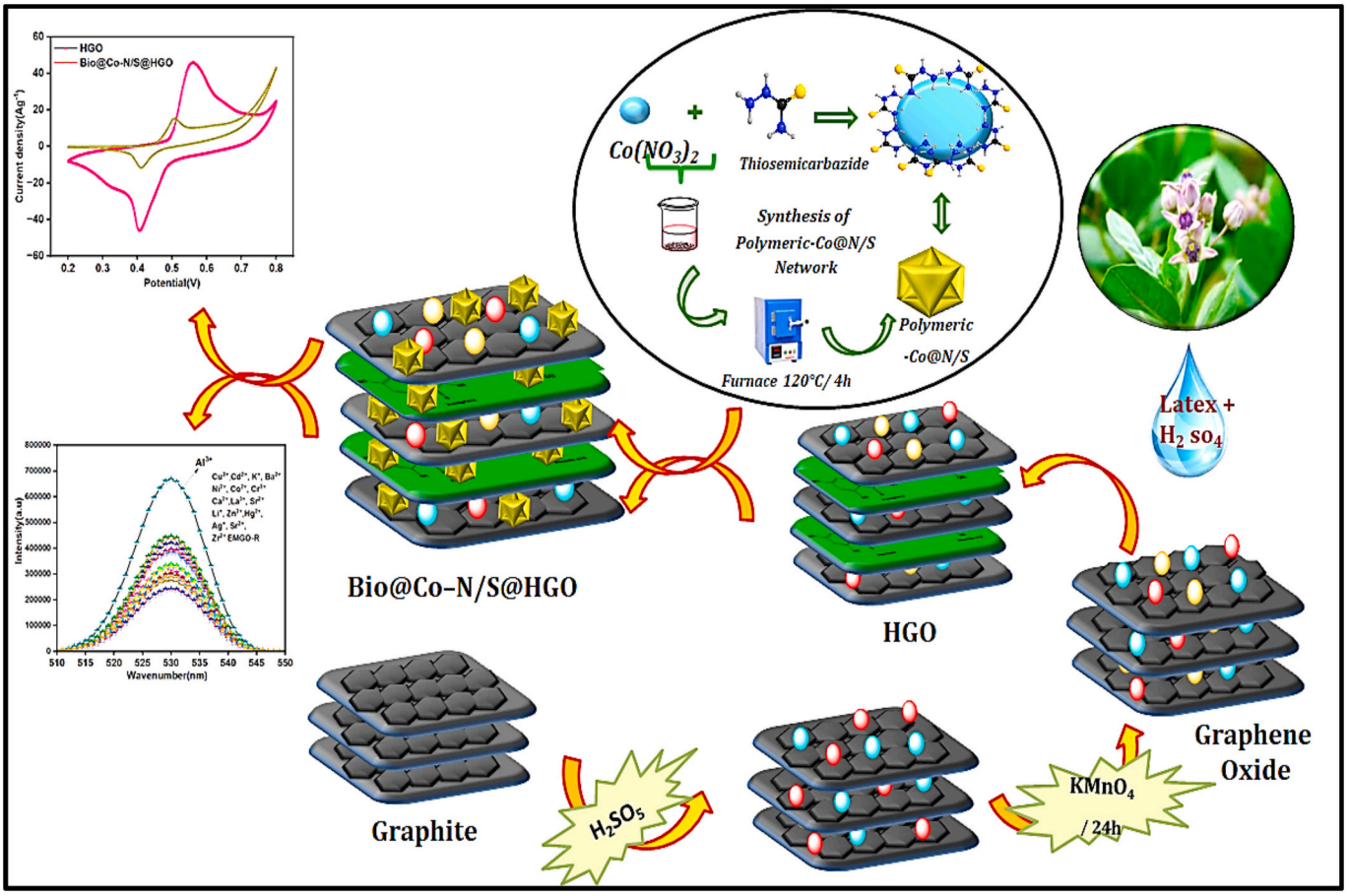
The latex derived from *Calotropis gigantea* was acquired from Valayapatti village, in the Namakkal district of Tamil Nadu, India. All aqueous solutions were prepared with deionized water. Graphite (500 mesh size), Cobalt nitrate hexahydrate, concentrated sulphuric acid (98.08%), and hydrogen peroxide (30%) from Merck Laboratory. Thiosemicarbazide and Potassium permanganate were sourced from Nice Chemicals. All reagents used in this study were of analytical grade**.**

2.2. Synthesis of bio functionalized graphene oxide (Bio@HGO)

Bio@HGO was prepared through a one-step oxidation–functionalization process, as illustrated in Scheme 1. Briefly, 0.5 g of graphite was dispersed in 40 mL of H_2SO_4 (98.5%) and stirred at room temperature for 30 min, followed by sonication for 60 min. The mixture was cooled to 0 °C, and 3 g of $KMnO_4$ was added gradually over 2 h, followed by stirring at room temperature for 24 h. Separately, 2 mL of *Calotropis gigantea* latex was mixed with 5 mL of concentrated H_2SO_4 and then added to the graphite oxide suspension. The mixture was stirred for another 3 h at room temperature. After cooling to 0 °C, 60 mL of deionized water was added carefully while maintaining the temperature below 10 °C, followed by the addition of 5 mL of 30% H_2O_2. The product was repeatedly washed with double-distilled water and centrifuged until the pH exceeded 5. Finally, the resulting brown Bio@HGO was oven-dried and stored in a desiccator. [28]

2.3. Synthesis of polymeric cobalt–TSC coordination network (Polymeric-Co@N/S-network)

Polymeric-Co@N/S-Network was synthesized through a solvothermal procedure. Typically, 1.6 g of thiosemicarbazide (TSC) was dissolved in 50 mL of a 1:1 ethanol–water mixtureand then subjected **to**sonication at °C to obtain a clear solution. Separately, **3**.2 g of cobalt nitrate hexahydrate ($Co(NO_3)_2{\cdot}6H_2O$)was dissolved in 30 mL of a 1:1 ethanol and deionized (DI) water mixture**.** The hot solution of TSC was then gradually introduce into to the cobalt precursor solution under constant stirring for about 60 min at room temperature.A notable color change from pink to purple was observed, indicating coordination between cobalt ions and thiosemicarbazide. The resulting mixture was subsequently transferred into a Teflon-lined stainless steel autoclave, sealed and heated at 120 °C for 2 h**.** After completion of the reaction, the autoclave was removed from the furnace, allowed to release the internal pressure, and then again placed in the furnace at 120 °C for an additional 2 h to ensure complete crystallization. Finally, the resulting precipitate was thoroughly washed with acetone**,** and dried in a hot air oven at 60 °C. The final product was obtained as dark purple crystals of bulk



Scheme 1. Schematic representation of the synthesis of HGO b) Synthesis of polymeric Cobalt–TSC coordination network c) Synthesis of Bio@Co–N/S@HGO

polymeric-Co@N/S-framework.

2.4. Synthesis of bio@Co–N/S@HGO

This process describes the preparation of a hybrid material through the integration of bio-functionalized graphene oxide (Bio@HGO) with bulk polymeric-Co@N/S-framework (metal–TSC coordination polymeric framework). Initially, 200 mg of synthesized HGO and polymeric-Co@N/S-framework were accurately weighed and dispersed separately in 30 mL of double-distilled water and ethanol, respectively. Both suspensions were ultrasonicated for 20 min to obtain a homogeneous and stable brown-colored colloidal solution (Bio@HGO), indicating complete exfoliation and dispersion, and a partially dispersed purple-colored suspension (polymeric-Co@N/S-framework), with slight sedimentation due to incomplete solubility. Subsequently, the HGO suspension was transferred to a round-bottom flask and magnetically stirred for 10 min to enhance dispersion stability. The previously prepared polymeric-Co@N/S-framework suspension was then added drop wise to the HGO suspension under continuous stirring, facilitating uniform interaction and possible interfacial bonding between the components, as evidenced by the development of a dark brown coloration. The resulting mixture was further stirred continuously for 2 h to ensure effective composite formation through physical adsorption and/or chemical interactions. After completion of the reaction, the suspension was washed several times with acetone to remove unreacted species and impurities. Finally, the product was collected and transferred to a petri dish for drying and subsequent characterization.

2.5. Characterization

A range of analytical techniques was employed to analyze the structural and physiochemical properties of GO and Bio@Co–N/S@HGO including crystalline size and shape; surface morphology; functional group composition; chemical composition; and defects. X-ray photoelectron spectroscopy (XPS) measurements were performed using a Carl Zeiss spectrosystem with monochromatic Al Kα radiation source (1486.6 eV). The instrument operated at t 15 kV and 15 mA, with spectra acquired in constant analyzer energy (CAE) mode using a pass energy of 50.0 eV, a 400 μm spot size, and an analyzer resolution of 4.2. The analysis chamber pressure was maintained at $\sim 5 \times 10^{-9}$ mbar to ensure high-vacuum conditions. Fourier-transform infrared spectroscopy (FT-IR) employed a Shimadzu IR-Prestige-21 instrument; X-ray diffraction (XRD) was analyzed using a Shimadzu XRD-6000 instrument; scanning electron microscopy (FE-SEM) analyses provided information on morphology; Raman spectroscopy employed a Horiba JobinYvon spectrometer with a 514 nm laser and was conducted within a controlled nitrogen environment. Fluorescence studies were carried out on a Jasco FP-8200 spectrofluorometer using 1 cm quartz cuvettes with both emission and excitation slit widths set to 5.0 nm. All emission spectra were observed at 24 ± 1 °C. Fresh stock solutions (2×10^{-3} M) of Bio@Co–N/S@HGO for analysis were freshly prepared in deionized water prior to analysis. The solutions of metal ions were prepared from the nitrate and chloride salts of Na^{+}, K^{+}, Al^{3+}, Ni^{2+}, Co^{2+}, Mn^{2+}, Ag^{+}, Ti^{2+}, Zn^{2+}, Pb^{2+}, Hg^{2+}, Cu^{2+}, La^{3+}, Cr^{3+}, and Fe^{3+}.

Electrochemical measurements were carried out using a symmetric two-electrode cell configuration with 1 M H_2SO_4 as the electrolyte. Two identical Bio@Co–N/S@HGO electrodes were used as the positive and negative electrodes, forming a Bio@Co–N/S@HGO//Bio@Co–N/

S@HGO symmetric supercapacitor. The electrodes were prepared by mixing the active material with poly(vinylidene fluoride) (PVDF) binder in a 9:1 mass ratio and coating the slurry onto glassy carbon substrates. The active material loading was maintained at 4.0 mg. The assembled symmetric device was used for cyclic voltammetry (CV), galvanostatic charge–discharge (GCD), and electrochemical impedance spectroscopy (EIS).

3. Result and discussion

Cobalt nitrate hexahydrate dissociates in the reaction medium to produce Co^{2+} ions, which coordinate with thiosemicarbazide (TSC) through the N- and S-donor atoms to form the Co–TSC coordination framework. Graphene oxide, containing oxygenated surface functionalities, provides a substrate for interaction and immobilization of the coordination component. *Calotropis gigantea* latex is introduced during the preparation of the graphene substrate as a bio-derived functionalization component. The organic constituents present in the latex may interact with the oxygen-containing groups of graphene oxide through hydrogen bonding, π–π interactions, and other interfacial interactions, resulting in modification of the surface chemistry and morphology. Subsequent integration of the Co–TSC coordination component produces the Bio@Co–N/S@HGO hybrid architecture. The observed structural and electrochemical characteristics therefore represent the combined effects of the functionalized graphene framework and the Co–N/S coordination component. Since a latex-free control was not investigated, the specific contribution of the latex-derived organic fraction to conductivity, stabilization, or charge-storage performance cannot be independently established.

X-ray diffraction (PXRD) analysis was performed to elucidate the structural characteristics, phase purity, and crystalline nature of GO, HGO and Bio@Co–N/S@HGO. The diffraction patterns of GO, HGO and Bio@Co–N/S@HGO are presented in Fig. 1. Pristine graphite exhibits a sharp diffraction peak at 26.5° corresponding to an interlayer d-spacing of 0.34 nm. Following oxidation to graphene oxide and subsequent modification to **Bio@Co–N/S@HGO**, a new characteristic diffraction peak appeared at $2\theta = 9.7^{o}$ for GO and 8.5° for Bio@Co–N/S@HGO. The shift to lower angles indicates a pronounced expansion of interlayer spacing from 0.3 to 1.6 nm. This is due to the introduction of oxygenated functionalities and incorporation of bioactive moieties derived from *Calotropis gigantea* latex and Co–TSC-containing crystalline component. Upon the incorporation of the Co–TSC-containing crystalline component from cobalt nitrate and thiosemicarbazide to create **Bio@Co–N/S@HGO** materials, additional sharp peaks in the low- and mid-angles were observed, confirming that a crystalline coordination framework has formed on the surface of the HGO material. The molecules representing these prominent peaks are found at approximately $2\theta = 11.2^{o}$, 16.8^{o}, 20.5^{o}, 21.4^{o}, 24.2^{o}, 25.2^{o}, 29.7^{o}, 32.2^{o}, 37.2^{o} and 43.3^{o} which are characteristic of the ordered Co–TSC-containing crystalline component structure, indicating the successful in situ growth and anchoring of the framework onto the graphene sheets. The peaks correspond well to the cobalt node's periodic arrangement which is coordinated by sulfur and nitrogen rich thiosemicarbazide ligands in a bulk crystalline structure. The characteristic diffraction peak of GO is much weaker or nearly absent in the composite of **Bio@Co–N/S@HGO** [29,30].

The morphological and micro structural features of GO, HGO and the **Bio@Co–N/S@HGO** composite were investigated with field emission scanning electron microscopy (FESEM) as illustrated in Fig. 2 (a, b). The morphologies of the pristine GO flakes were typical wrinkly, crumpled, stacked; showing a thin sheet-like structure indicating their high specific surface area and flexibility. Fig. 2(c, d) Bio-functionalizing with Calotropisgigantea latex created a much rougher and textured surface on the HGO sheets compared to the GO flakes confirming the successful adsorption and coating of biomolecules derived from the latex. The different morphologies resulted from the deposition of natural polymers and bioactive constituents from the latex that interact with GO through hydrogen bond formation, van der Waals forces, and strong π–π stacking. The incorporation of modified Polymeric-Co@N/S crystalline into the HGO systems created a new set of morphological characteristics. The Fig. 2(e, f) SEM images of the **Bio@Co–N/S@HGO** composite confirm that Polymeric-Co@N/S crystallites are uniformly present in nano-sized crystalline domains attached to the graphene sheets with very little

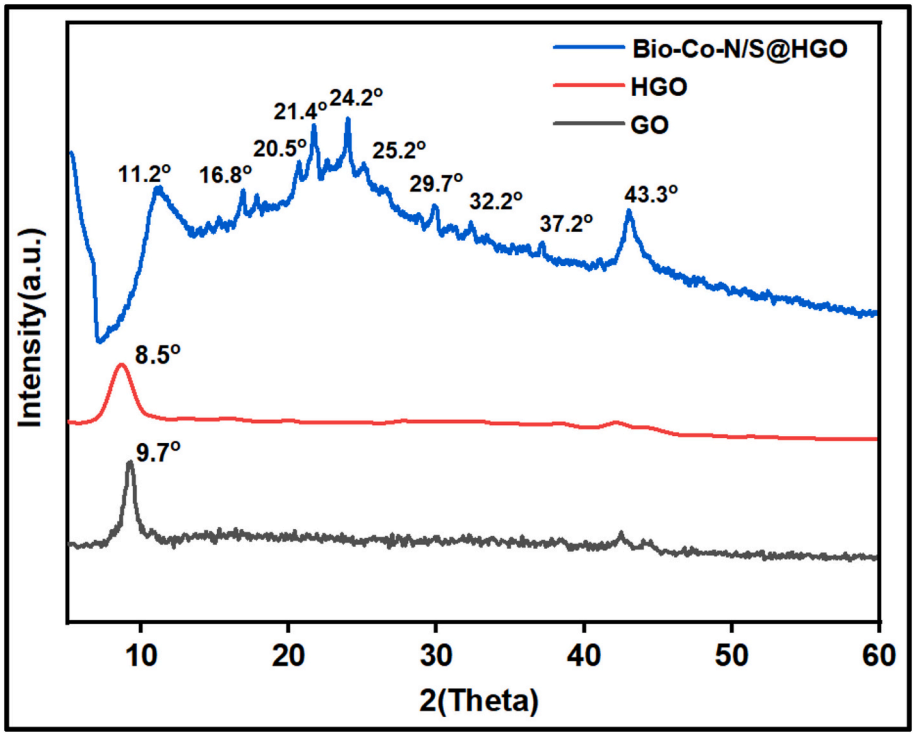


Fig. 1. XRD analysis of graphite, GO, HGO and Bio@Co-N/S@HGO.

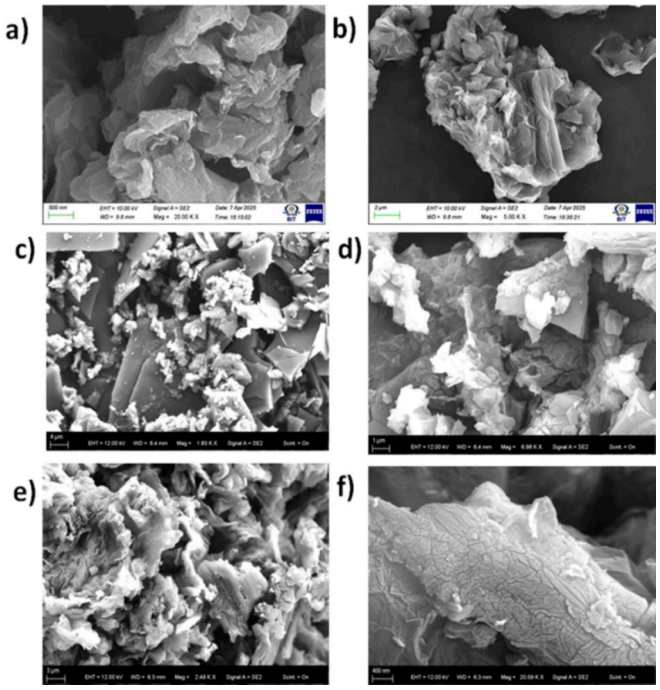


Fig. 2. FE-SEM images of (a, b)H-GO, (c, d) Polymeric-Co@N/S-Network,(e, f) Bio@Co–N/S@HGO.

agglomeration. This demonstrates that there were strong interfacial interactions between the functional groups of **Bio@Co–N/S@HGO**, which have oxygen and nitrogen, and cobalt ions during nucleation of the Polymeric-Co@N/S framework, causing good distribution across the surface area. This morphology results in increased surface roughness, greater exposure of electroactive sites, and better electrolyte penetration & ion diffusion pathways [31].

Raman spectroscopy was employed to evaluate the structural disorder and defect characteristics of HGO and Bio@Co–N/S@HGO, as shown in Fig. 3. Both samples exhibit the characteristic D and G bands of the graphene framework, which are associated with defect-induced disorder and the in-plane vibration of sp^2 carbon, respectively. For HGO, the D and G bands are located at 1351 and 1596 cm^{-1}, respectively, with an ID/IGI_D/I_G ratio of 0.98. Following incorporation of the Polymeric-Co@N/S framework, the D and G bands of Bio@Co–N/S@HGO appear at 1351 and 1587 cm^{-1}, respectively, with a slightly increased ID/IGI_D/I_G ratio of 1.03. This variation indicates a modest change in the defect density and local structural disorder of the graphene framework after composite formation. The observed band shifts and change in ID/IGI_D/I_G suggest structural perturbation associated with the incorporation of the Polymeric-Co@N/S framework. However, these Raman features alone do not establish the formation of specific covalent bonds or a substantial increase in sp^3 carbon content. The chemical composition and interfacial interactions of the composite are further supported by the complementary FT-IR and XPS analyses [32,33]. (See Fig. 5.)

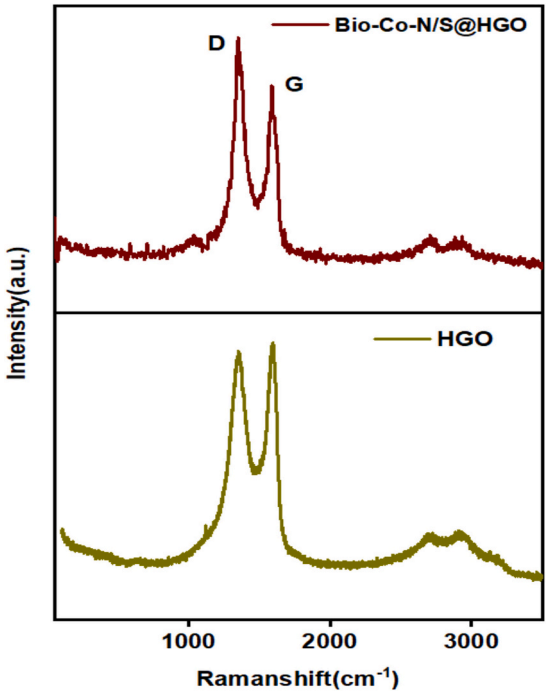


Fig. 3. Raman spectrum of HGO and Bio@Co–N/S@HGO.

X-ray photoelectron spectroscopy (XPS) analysis was performed to evaluate the elemental composition and chemical states of bio functionalized HGO and Bio@Co–N/S@HGO. GOwas developed using latex secreted by *Calotropis gigantea*, while **Bio@Co–N/S@HGO** was produced by grafting a metal organic coordination formation from cobalt nitrate and thiosemicarbazide onto the surface of HGO (Fig. 4 a). The survey spectrum of HGO and Bio@Co–N/S@HGO exhibits sharp and intense peaks of C1s ≈ 284 eV, O1s ≈ 532 eV peak along with additional N1s ≈ 399 and S2p ≈ 164 signals, which confirm the incorporation of nitrogen and sulfur containing functional groups were successfully decorated surface of graphene surface. An additional Fig. 4b metal-related peak observed in Bio@Co–N/S@HGO indicates the coordination of the cobalt species (≈780.6 & 782.8) within the composite [27,34]. High-resolution XPS was employed to determine the chemical composition of the synthesized Bio@Co–N/S@HGO,. These elemental data are presented in Table 1 and provide confirmation of elements (C/O/N/S/Co) as being successfully incorporated during the process of functionalization.

The high-resolution XPS spectra reveal multiple chemical environments associated with the surface functional groups of the Bio@Co–N/S@HGO composite. As shown in Fig. 5(b), the deconvoluted C 1s spectrum of Bio@Co–N/S@HGO consists of five components at 284.2 eV (C=C; sp^2 carbon), 285.5 eV (C–N/C–S), 286.3 eV (C–OH), 287.6 eV (C–O–C), and 288.3 eV (O–C=O). The deconvoluted N 1s spectrum in Fig. 5(c) shows peaks at 399.8 and 401.3 eV, attributed to amine- and amide-type nitrogen species, respectively. Furthermore, the S 2p spectrum in Fig. 5(d) exhibits peaks in the 163–165 eV region, indicating the presence of sulfur-containing chemical environments and supporting the functionalization of the composite. In the deconvoluted C 1s spectrum of Bio@HGO, C–N and C–S contributions are observed alongside oxygen-containing species, suggesting enhanced surface functionalization. This modification is also reflected in the reduced relative intensity of sp^2 carbon compared with pristine GO. As shown in Fig. 5(a), the high-resolution C 1s spectrum of pristine GO can be deconvoluted into four components at approximately 284.0, 286.1, 287.6, and 288.0 eV, corresponding to C–C/C=C, C–O, C=O, and O–C=O groups, respectively. Overall, the spectral differences among GO, Bio@HGO, and Bio@Co–N/S@HGO support progressive surface functionalization and the incorporation of nitrogen-, sulfur-, and cobalt-containing components into the graphene-based hybrid [35,36].

It should be noted that the present characterization provides evidence for the combined effects of latex-derived surface modification and Co–TSC incorporation rather than quantitatively separating their individual contributions. The changes observed in the FT-IR and XPS analyses support latex-associated surface functionalization, while the subsequent incorporation of Co–TSC introduces additional Co, N and S-containing environments. Thus, the structural and physicochemical properties of Bio@Co–N/S@HGO are attributed to the combined effects of bio-derived functionalization, Co–N/S coordination, and the

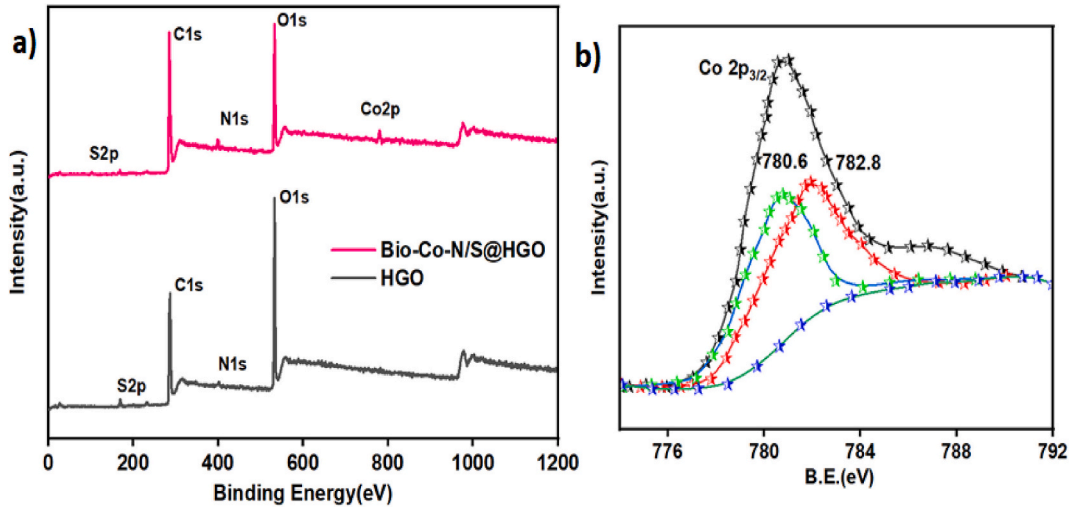


Fig. 4. XPS survey spectrum of HGO and Bio@Co–N/S@HGO b) deconvoluted Co spectra.

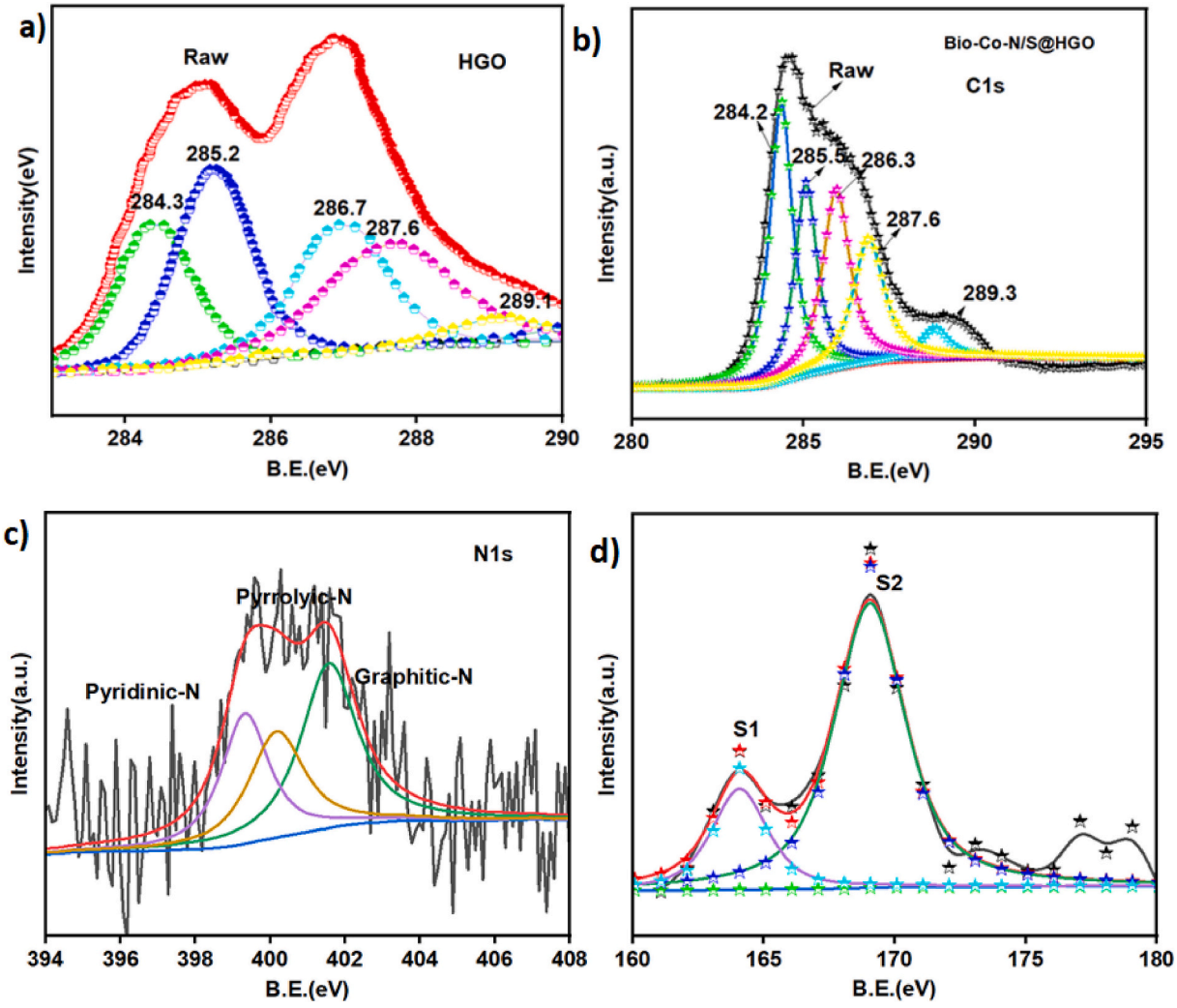


Fig. 5. XPS for a) deconvoluted C1s spectra for HGO b) deconvoluted Bio@Co–N/S@HGO spectra of C1s C) Deconvoluted N1s spectra d) deconvoluted S2p spectra of Bio@Co–N/S@HGO.

graphene framework. In the absence of a latex-free control, the specific contribution of latex to conductivity, stabilization, and electrochemical performance cannot be independently quantified.

The functionalization of graphene oxide using bioactive moieties from *Calotropis gigantea* latex and the subsequent formation of the Bio@Co–N/S@HGO composite were confirmed by FTIR analysis, as

Table 1
Surface elemental composition of HGO and **Bio@Co–N/S@HGO** from XPS analysis.

Sample name	Carbon (Wt%)	Oxygen (Wt%)	Nitrogen (Wt%)	Sulfur (Wt%)	Cobalt (Wt%)
HGO	53.3	37.9	5.4	3.4	–
Bio@Co–N/S@HGO	56.1	27.8	6.7	5.1	6.2

shown in Fig. 6. FTIR analysis of pristine GO exhibits characteristic absorption peaks observed at 1042, 1672, 1749, and 3484 cm $-^{1}$ for C—O stretching of epoxide groups, aromatic C═C skeletal vibrations, carbonyl (C=O) stretching, and hydroxyl (O—H) stretching, confirms the presence of various oxygenated functionalities. After the bio-functionalization, the FTIR spectrum of the HGO exhibited a new adsorption peak that represents the nitrogen functional groups, suggesting the interaction of latex-derived biomolecules with the GO surface. In addition, new peaks for C—N stretch and amide, and carbonyl functional groups demonstrating the nucleophilic epoxy ring-opening reaction and the formation of covalent C—N bonds. Upon further incorporation of the modified Polymeric-Co@N/S crystalline, the Bio@Co–N/S@HGO composite shows some additional distinct peaks which was observed at 1320 cm^{-1}, 1641 cm^{-1}, 1734 cm^{-1}corresponding to the C—N stretching, amide C═O stretching vibrations, and non-conjugated C═O (of carboxylic acids or esters), respectively.

The broad peak observed at 3220 $cm^{-1}$and 1039 cm^{-1}correspond to O—H and NH stretching vibrations, indicative of hydroxyl and amine functionalities generated during epoxy ring cleavage. Additionally, the weak feature near 2330 cm^{-1} may be associated with adsorbed CO_2 or atmospheric interference. These spectral features confirm the successful incorporation of nitrogen-containing groups and the effective bio-functionalization of the GO framework. The decrease in the C—O stretching peak (shift) from 1042 cm^{-1} for GO to lower wave numbers for Bio@Co–N/S@HGO shows the chemical modification of the epoxy groups during the bio-functionalization of GO and subsequent incorporation of the Polymeric-Co@N/S framework to produce new C—N and related covalent bonds. Thus, the observed changes in spectra provide evidence of bio-functionalization of GO and successful incorporation of the Polymeric-Co@N/S framework. [37,38]

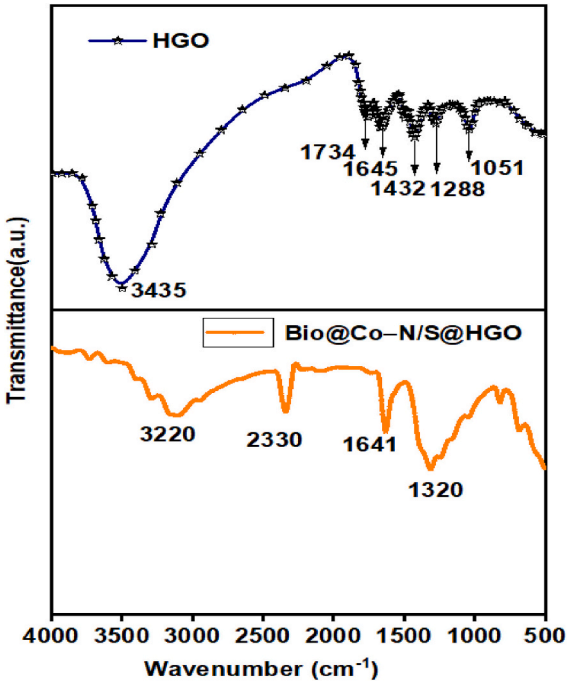


Fig. 6. FT-IR analysis of HGO and Bio@Co–N/S@HGO.

4. Electrochemical performance study

The introduction of heteroatoms functionalities and porous frameworks into graphene oxide significantly improves its electrochemical energy storage characteristics. To assess the supercapacitive behavior of pristine HGO and the Bio@Co–N/S@HGO composite, their electrochemical properties were comprehensively analyzed using cyclic voltammetry (CV), galvanostatic charge–discharge (GCD), and electrochemical impedance spectroscopy (EIS).

The electrochemical performance of pristine HGO and the Bio@Co–N/S@HGO composite were evaluated through cyclic voltammetry (CV) within a potential range of 0.2–0.8 V using 1 M H_2SO_4 as an electrolyte. All measurements were performed in a symmetric two-electrode configuration in which two identical Bio@Co–N/S@HGO electrodes were employed as the positive and negative electrodes. Fig. 7(a) compares the CV profiles of HGO and Bio@Co–N/S@HGO at a scan rate of 10 mV s^{-1}. The composite electrode displays more expanded CV area and higher current density than pristine HGO, indicating enhanced charge storage capacity. The appearance of distinct redox peaks confirms the contribution of faradaic reactions, arising from the reversible Co^{2+}/Co^{3+} redox transitions and heteroatom (N/S)-induced active sites, demonstrating a dominant pseudocapacitive behavior [39,40]. Fig. 7(b) presents the CV profiles of Bio@Co–N/S@HGO at different scan rates (10–100 mV s^{-1}). The quasi-rectangular shape with distinct redox peaks features even at increased scan rates, highlights excellent rate capability and efficient ion transport within the electrode material. The increase in current response with scan rate further confirms good electrochemical reversibility and fast charge-transfer kinetics. For comparison, Fig. 7(c) presents the CV analysis of HGO at various scan rates. Although HGO shows similar redox features,

The current response and integrated area are significantly lower than those of the composite, indicating limited electrochemical activity due to fewer active sites and lower conductivity. The change in specific capacitance with scan rate is depicted in Fig. 7(d). The Bio@Co–N/S@HGO symmetric device exhibited a CV-derived specific capacitance of 598.4 F g^{-1} at 10 mV s^{-1}, which gradually decreases to ~244 F g^{-1} at 100 mV s^{-1}. The capacitance decay at higher scan rates is attributed to limited diffusion of electrolyte ions to access inner active sites. Nevertheless, the composite maintains superior capacitance retention, highlighting its efficient ion transport and structural stability. The enhanced electrochemical response of Bio@Co–N/S@HGO is associated with the combined characteristics of the Co–N/S coordination environment, cobalt redox activity, heteroatom-containing surface functionalities, and the interconnected graphene-based architecture. These structural features can provide accessible electroactive sites and facilitate electrolyte-ion transport within the hybrid electrode. The present data demonstrate the electrochemical behavior of the complete Bio@Co–N/S@HGO architecture, however, because a latex-free analogue was not examined, the individual contribution of the latex-derived organic fraction to the measured capacitance and charge-transfer behavior cannot be quantitatively separated.

To further elucidate the charge-storage mechanism of Bio@Co–N-S@HGO, Dunn's method was employed to separate the surface-controlled and diffusion-controlled contributions from the CV curves recorded at scan rates of 10, 20, 30, 40, 50, and 100 mV s^{-1}. The current response was analyzed using the relationship $i(V) = k_1\nu^{1/2} + k_2\nu$, where $k_1\nu^{1/2}$ and $k_2\nu$ represent the diffusion-controlled and surface-controlled (capacitive) contributions, respectively. The capacitive contribution increased progressively from 24.77% at 10 mV s^{-1} to 51.01% at 100 mV s^{-1}, while the corresponding diffusion-controlled contribution decreased from 75.23% to 48.99%. This variation demonstrates the coexistence of surface-controlled and diffusion-controlled charge-storage processes. The dominant diffusion contribution at lower scan rates indicates enhanced electrolyte-ion penetration and Faradaic reaction

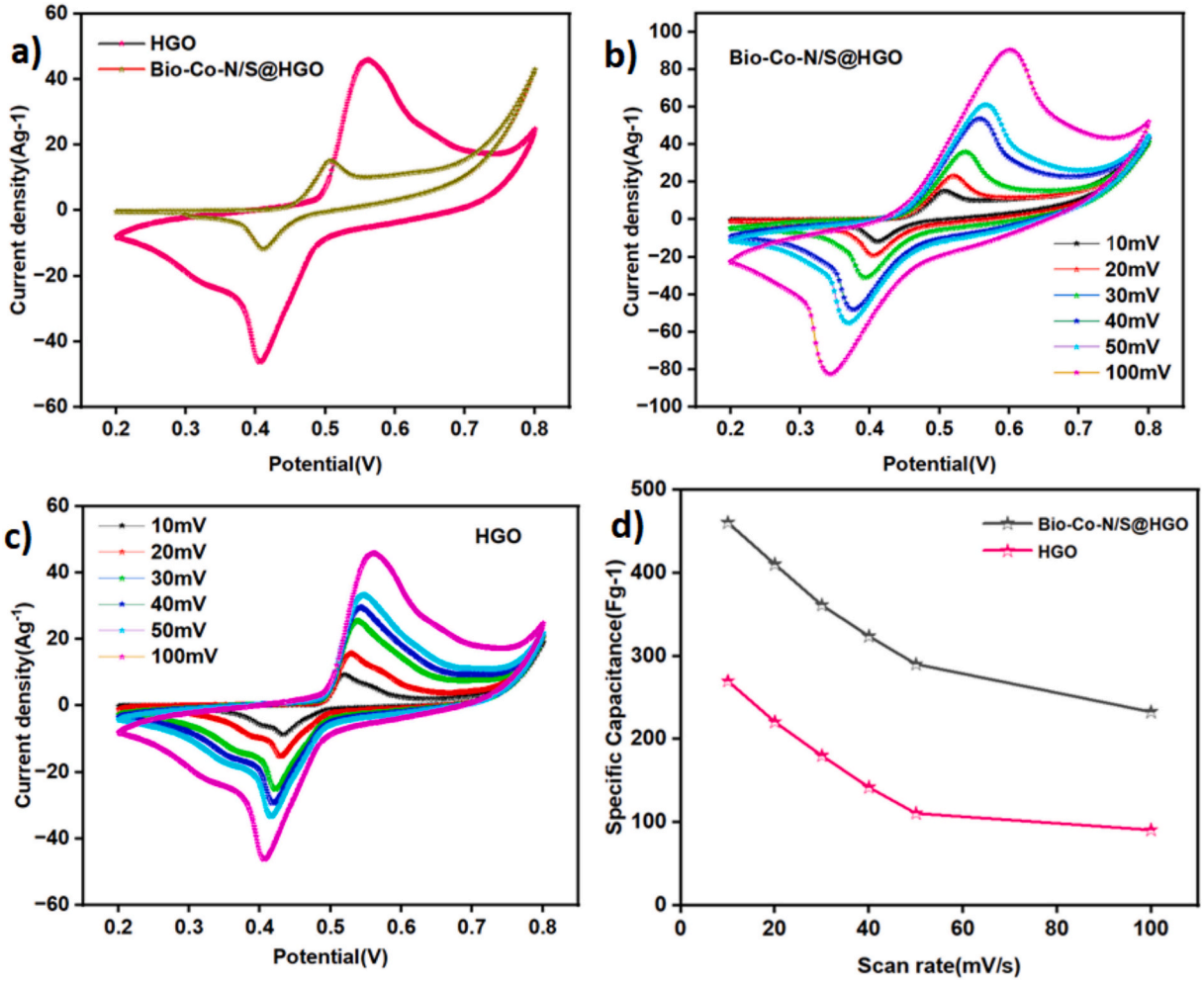


Fig. 7. Presents the electrochemical analysis of the electrode materials in a two electrode system using 1 M H_2SO_4 (a) CV curves of HGO and Bio@Co–N/S@HGO at a scan rate of 10 mV s^{-1}, (b) CV spectra of Bio@Co–N/S@HGO at varying scan rates, (c) CV spectra of HGO at varying scan rates, (d) comparison of specific capacitance between HGO and Bio@Co–N/S@HGO.

kinetics, whereas the increasing capacitive contribution at higher scan rates is associated with rapid surface-controlled charge storage. These findings, together with the b-value analysis, confirm the mixed kinetic nature of Bio@Co–N-S@HGO and provide quantitative support for its pseudocapacitive charge-storage behavior [41].

The galvanostatic charge–discharge (GCD) performance of the Bio@Co–N/S@HGO//Bio@Co–N/S@HGO symmetric supercapacitor was investigated at current densities of 1, 2, 3, 5, 7, and 10 A g^{-1} within an operating voltage window of 0–0.60 V. As the current density increased, the discharge time gradually decreased, indicating the expected high-rate charge–discharge behavior of the symmetric device. The device-specific capacitance was calculated using the total active mass of both electrodes (4.0 mg). Furthermore, the energy density and power density were determined from the experimentally obtained discharge curves. The symmetric Bio@Co–N/S@HGO device delivered energy densities ranging from 2.23 to 2.97 Wh kg^{-1} and corresponding power densities of 150.1–1502.2 W kg^{-1}, demonstrating promising energy-storage capability together with efficient power delivery [42–44].

For a symmetric two electrode device, using the total activemass of both electrodes:

$$C_{device} = I\Delta t/(m_+ + m_-)\Delta V \quad (1)$$

- I = discharge current (A), Δt = discharge time (s), m_+ and m_- = active masses of positive and negative electrodes, ΔV = discharge voltage window (V)

$$E = \frac{1}{2}C_{device}(\Delta V)^2 2 \Big/ 3.6 \quad (2)$$

Where E is energy density in Wh kg^{-1}

$$P = E \times 3600/\Delta t \quad (3)$$

Where P is power desity in W kg^{-1}.

Electrochemical impedance spectroscopy (EIS) was employed to investigate the resistance characteristics, charge-transfer response, and ion-transport behavior of HGO and Bio@Co–N/S@HGO in the assembled electrochemical system. The Nyquist plots (Fig. 8c) display a typical profile with a depressed semicircle in the high-frequency region and an inclined response in the low-frequency region, corresponding to interfacial charge-transfer processes and ion-transport/diffusion behavior, respectively. The intercept on the real axis represents the equivalent series resistance (Rs), which includes contributions from the electrolyte, electrode, current-collector/substrate contact, and other components of the assembled cell. Bio@Co–N/S@HGO exhibits a lower Rs value than HGO, indicating lower overall series resistance in the assembled electrode system. Furthermore, the smaller semicircle diameter observed for the composite suggests a reduced charge-transfer resistance (Rct) and a

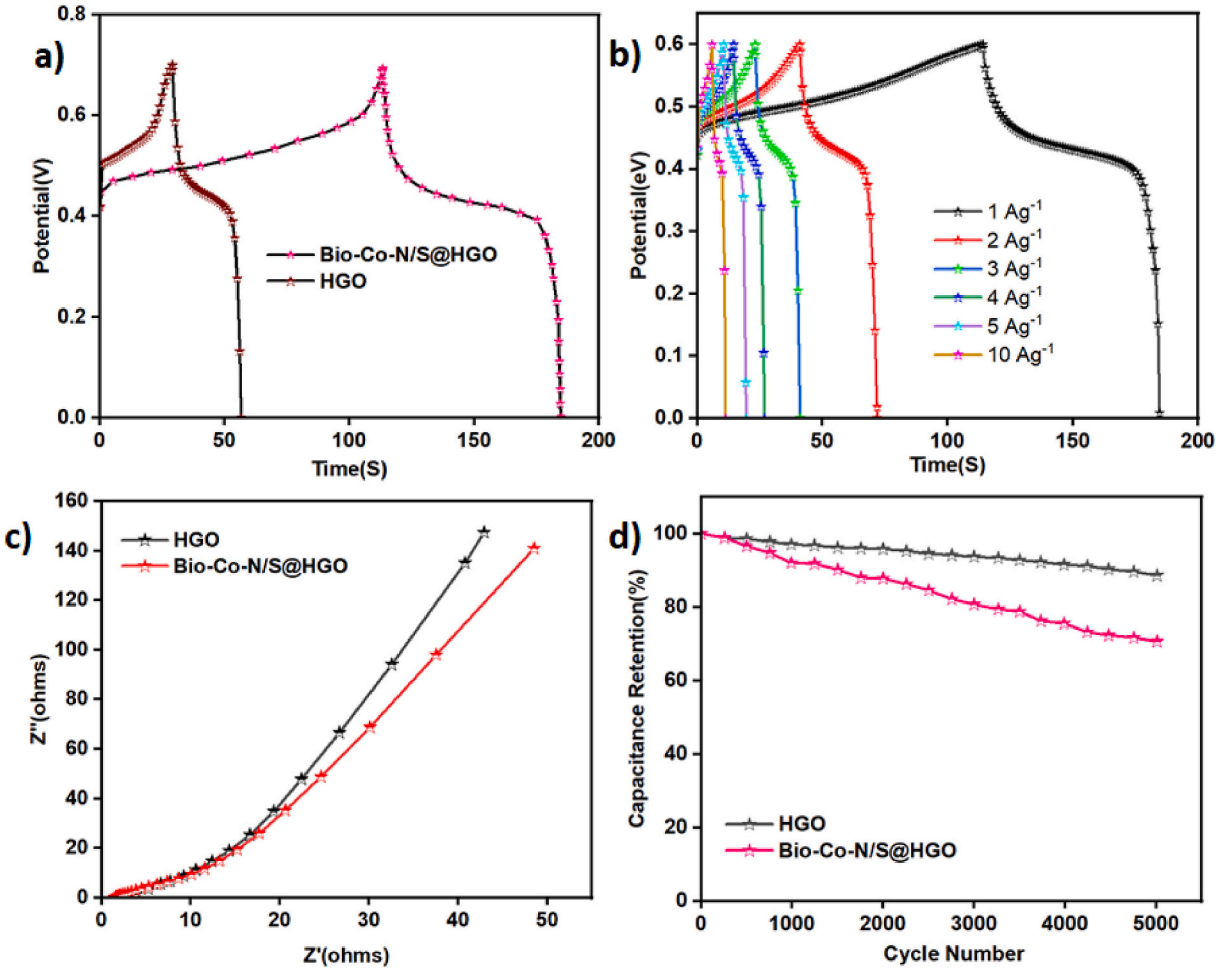


Fig. 8. Electrochemical performance of the Bio@Co–N/S@HGO//Bio@Co–N/S@HGO symmetric supercapacitor: (a) comparison of GCD profiles of HGO and Bio@Co–N/S@HGO at 1 A g^{-1}, (b) GCD profiles of the Bio@Co–N/S@HGO symmetric device at different current densities, (c) Nyquist plots of HGO and Bio@Co–N/S@HGO, and (d) cycling stability and capacitance retention over 5000 charge–discharge cycles.

more favorable interfacial charge-transfer response. In the low-frequency region, the steeper slope approaching a vertical response indicates more pronounced capacitive behavior and favorable ion-transport characteristics. These EIS features are consistent with the CV and GCD results and support the improved electrochemical response of the Bio@Co–N/S@HGO-based symmetric device. However, the EIS measurements reflect the combined contributions of the electrolyte, electrode interfaces, contacts, and porous-electrode transport and therefore do not provide a direct measurement of the intrinsic electrical conductivity of the material [45,46].

The long-term electrochemical stability of HGO and Bio@Co–N/S@HGO electrodes was evaluated over 5000 charge–discharge cycles, as presented in Fig. 8(d). The Bio@Co–N/S@HGO electrode exhibits superior cycling stability, retaining approximately ~94% of its initial capacitance after 5000 cycles. The enhanced stability of the Bio@Co–N/S@HGO composite arises from the robust structural integration between cobalt active centers and the N/S-functionalized graphene framework, which provides structural robustness and prevents material degradation during repeated cycling. The long-term cycling stability of Bio@Co–N/S@HGO is associated with the structural integration of the cobalt coordination component with the N/S-functionalized graphene framework. The interconnected architecture and heteroatom-containing surface functionalities may contribute to maintaining accessible electroactive sites and stable electrode–electrolyte interfaces during repeated cycling. Nevertheless, the individual contribution of the latex-derived organic fraction to the observed cycling stability cannot be independently quantified from the present experiments. A comparison of the electrochemical performance of Bio@Co–N/S@HGO with previously reported graphene-based electrode materials is presented in Table 2, highlighting its specific capacitance and cycling stability under the respective experimental conditions.

5. Metal ion selectivity studies

The interaction between GO and the Bio@Co–N/S@HGO nanocomposite was examined by UV–visible spectroscopic analysis. The absorption behavior of both GO and the Bio@Co–N/S@HGO system was evaluated at identical concentrations for direct comparison. The GO sample exhibited two characteristic absorption bands centered at approximately 235 and 296 nm. In contrast, the Bio@Co–N/S@HGO chemosensor showed a noticeably altered spectral profile, indicating a clear modification in the optical response upon composite formation. These differences in absorption features suggest changes in the electronic structure and interfacial interactions between GO and the incorporated Co–N/S functional framework. [53]

Fluorescence emission spectroscopy was used to investigate the photophysical properties of the prepared materials. The emission profiles presented in Fig. 9a enable a direct comparison between the individual components and the resulting nanocomposite, providing insight into their fluorescence characteristics. Notably, the Bio@Co–N/S@HGO system exhibits a discernible shift in emission behavior relative to GO and related counterparts. This change is commonly attributed to

Table 2
Comparison of the electrochemical performance of Bio@Co–N/S@HGO nanosheets with previously reported graphene-based materials functionalized with long-chain organic molecules possessing similar functional groups.

S·No	Electrode material	Electrolyte	Cell configuration	Capacitance / test condition	Voltage window	Active mass	Cycling condition	Ref.
1	GO-bis(PIEA)	6 M KOH	Three-electrode	287 F g^{-1} at 10 mV s^{-1}	−0.4–0.4 V	0.5 mg	94.5% after 1000 cycles	[34]
2	FGO-TBpa	6 M KOH	Three-electrode	305 F g^{-1} F g^{-1} at 10 mV s^{-1}	0.0–0.4 V	0.5 mg	90% after 1000 cycles	[38]
3	TB-GO/DEA-GO	1 M H_2SO_4	Three-electrode	252 F g^{-1} at 10 mV s^{-1}	0.0–0.4	0.5 mg	90% after 1000 cycles	[2]
4	SAS-G	1 M H_2SO_4	Three-electrode	170 F g^{-1} at 4.5 A g^{-1} (GCD)	−0.4- 0.6 V	0.6 mg	91% after 1000 cycles	[47]
5	Salen-fGO	1 M H_2SO_4	Three-electrode	402.5 F g^{-1} at 10 mV s^{-1}	0.0–0.4 V.	0.5 mg	95% after 1000 cycles	[48]
6	ADS-G	1 M H_2SO_4	Three-electrode	210 F g^{-1} at 4.5 A g^{-1} (GCD)	0.0–0.6 V	0.4 mg	90% after 1000 cycles	[49]
7	PSSS-GR	1 M H_2SO_4	Two-electrode comparison	210 F g^{-1} at 5 A g^{-1} (GCD)	−0.4-0.8 V	0.4 mg	95% after 1000 cycles	[50]
8	Poly(*p*-phenylenediamine)–graphene	1 M H_2SO_4	Two-electrode symmetric	248 F g^{-1} at 2 A g^{-1} (GCD)	−0.4 – 0.4 V	0.4 mg	72% after 1000 cycles	[51]
9	RGO–PPy	3 M KCl	Two-electrode symmetric	224 F g^{-1} at 1 **A g^{-1}** (GCD)	-0.4-0.8 V	0.5 mg	83% after 5000 cycles	[52]
10	**Bio@Co–N/S@HGO**	**1 M H_2SO_4**	**Two-electrode symmetric**	**598.4 F g^{-1} at 10 mV s^{-1} (CV)**	**0.2–0.8 V**	**4.0 mg total**	**~94% after 5000 GCD cycles**	**Present work**

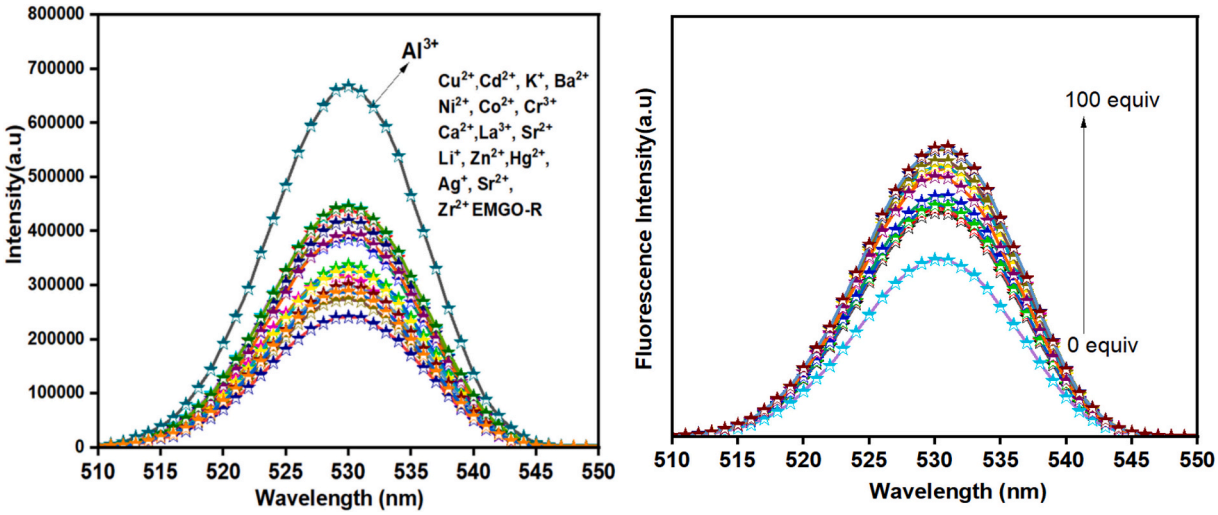


Fig. 9. (a) Selectivity study of metal ions, (b) Fluorescence response with various equivalents of analyte.

photoinduced electron transfer processes within the nanocomposite, which typically lead to fluorescence quenching. Interestingly, an enhancement in fluorescence response is observed, which can be rationalized by the formation of intermolecular hydrogen bonding interactions between nitrogen-containing functional groups of Bio@Co–N/S@HGO and poly moieties sites on GO. These findings indicate a strong interfacial interaction between graphene oxide and the Bio@Co–N/S@HGO framework, facilitating the formation of a stable and functionally active organic nanocomposite system.

The bio-derived hybrid graphene oxide material (Bio@Co–N/S@HGO), prepared using *Calotropis gigantea* latex, was explored as a fluorescent probe for the selective recognition of various metal ions, particularly those of environmental and biomedical concern. The interaction between Bio@Co–N/S@HGO and a range of metal ions, including Fe^{3+}, Cu^{2+}, Zn^{2+}, Ni^{2+}, La^{3+}, Bi^{2+}, Pb^{2+}, Zr^{2+}, V^{4+}, Cd^{2+}, Ti^{4+}, Hg^{2+}, Al^{3+}, Co^{2+}, Cr^{3+}, Sr^{2+}, Ag^{+} and Mn^{2+} was systematically investigated in aqueous HEPES buffer (50 mM, pH 7.7) using fluorescence spectroscopy (Fig. 9a). Emission studies were performed with an excitation wavelength of 260 nm, corresponding to the absorption profile. A pronounced increase in the intensity at 530 nm was observed upon the introduction of 100 equivalents of Al^{3+} ions to the sensor system. [54,55]

Fluorescence emission spectroscopy was used to investigate the response of Bio@Co–N/S@HGO toward different metal ions. As shown in Fig. 9a, the material exhibits an emission band centered at approximately 530 nm, with Al^{3+} producing a pronounced fluorescence enhancement compared with the other investigated ions. The fluorescence titration further showed a progressive increase in emission intensity at around 530 nm with increasing Al^{3+} concentration from 0 to 100 equivalents, followed by a tendency toward saturation at higher concentrations (Fig. 9b). At the selected pH of 7.7, Al^{3+} hydrolysis and speciation occur under the aqueous sensing conditions. Therefore, the observed fluorescence enhancement is more appropriately interpreted as an Al-associated interaction involving Al^{3+} and hydrolyzed Al species with the N/O-containing functional sites of Bio@Co–N/S@HGO. Such interactions may alter the photoinduced electron-transfer pathways and consequently enhance the fluorescence response. Although the selective fluorescence response toward Al-containing species is supported by the metal-ion selectivity and titration experiments, contributions from hydrolyzed species, aggregation, or precipitation cannot be completely excluded based solely on the fluorescence measurements. Accordingly, the sensing mechanism is interpreted conservatively as an Al^{3+}-

associated fluorescence enhancement rather than definitive evidence of a single molecular coordination mechanism [56,57].

The fluorescence titration data were quantitatively analyzed using the emission intensity at 530 nm. A progressive increase in fluorescence intensity was observed with increasing Al^{3+} concentration, with a linear response over 0.2–2.0 equivalents of Al^{3+}. The corresponding calibration equation was $F_{530} = 7.5721 \times 10^5 C + 4.0650 \times 10$ [6], with $R^2 = 0.9694$, where F_{530} represents the fluorescence intensity at 530 nm and C represents the Al^{3+} concentration in equivalents. The calibration slope of 7.5721×10^5 a.u. equivalent^{-1} was considered the analytical sensitivity. Ten independent blank measurements at 530 nm gave a mean fluorescence intensity of 423,568.41 a.u. with a standard deviation of 1262.45 a.u. and an RSD of 0.298%. Using the blank-based expressions LOD = 3.3σ/m and LOQ = 10σ/m, where σ is the standard deviation of the blank measurements and m is the calibration slope, the LOD and LOQ were calculated as 0.00550 and 0.01667 equivalents of Al^{3+}, respectively [58].

5.1. *Interference study of Al^{3+} fluorescence response in the presence of competing metal ions*

In contrast, the presence of other metal ions under identical experimental conditions resulted in negligible changes in emission behavior, indicating a preferential fluorescence response toward Al^{3+} under the investigated conditions. Competitive binding experiments were carried out to assess the influence of coexisting metal ions on Al^{3+} recognition. As shown in Fig. 10(a), the fluorescence response of Bio@Co–N/S@HGO was recorded upon simultaneous addition of 100 equivalents of Al^{3+} along with equal amounts of potentially competing metal cations. This observation indicates a preferential fluorescence response toward Al^{3+} under the investigated experimental conditions, with negligible interference from the tested competing ions. However, further quantitative selectivity evaluation using realistic competing-ion concentrations and mixed-ion matrices would be required to establish the practical performance of the sensor in complex real samples. The emission intensity associated with Al^{3+} remained essentially unaffected in the presence of these ions, indicating negligible interference. These results indicate a preferential fluorescence response toward Al^{3+} under the investigated experimental conditions, Further quantitative selectivity studies using realistic competing-ion concentrations and mixed-ion matrices are required to assess its applicability to complex samples.

The pH-dependent photophysical properties of Bio@Co–N/S@HGO in the presence of Al^{3+} were systematically examined over a pH range of 2–12 in aqueous medium (Fig. 10b). Fluorescence emission spectra were performed at a fixed excitation wavelength of 260 nm to monitor emission variations. As shown in the figure, the Bio@Co–N/S@HGO system in the presence of Al^{3+} exhibits a significant fluorescence enhancement in fluorescence intensity compared to the pristine material across the studied pH range. The fluorescence intensity gradually increases from acidic conditions and reaches a maximum in the neutral to slightly alkaline region (pH 7–8), followed by noticeable decrease at higher pH values. The enhanced emission observed in the presence of Al^{3+} may be associated with changes in the protonation–deprotonation equilibrium of nitrogen-containing functionalities and/or interactions of Al^{3+} or hydrolyzed Al species with the surface functional groups. Notably, a stable and intensified fluorescence response is maintained within the pH range of 7–10, indicating the highest fluorescence response under the investigated conditions. Accordingly, pH 7.7 was selected as the working condition for subsequent fluorescence sensing studies because it provided the maximum fluorescence response under the investigated conditions. [59]

6. Conclusion

A sustainable bio-assisted strategy was developed for the construction of a Co–N/S coordination hybrid integrated with functionalized graphene oxide (Bio@Co–N/S@HGO). In the present synthesis, *Calotropis gigantea* latex was employed as a bio-derived functionalization component during the preparation of the graphene substrate, followed by integration with the cobalt–thiosemicarbazide coordination framework. Structural and spectroscopic analyses confirmed the formation of the hybrid architecture containing Co, N, and S functionalities. The resulting Bio@Co–N/S@HGO electrode exhibited a specific capacitance of 598.4 F g^{-1} at 10 mV s^{-1} and approximately 94% capacitance retention after 5000 charge–discharge cycles. The symmetric device delivered energy densities of 2.23–2.97 Wh kg^{-1} and corresponding power densities of 150.1–1502.2 W kg^{-1} over the investigated current-density range. These electrochemical characteristics are discussed in relation to the combined effects of the Co–N/S coordination environment, cobalt redox activity, heteroatom-containing functional groups, and the graphene-based interconnected architecture. In addition, the hybrid exhibited an Al^{3+}-associated fluorescence turn-on response under the investigated conditions. The response showed a preferential enhancement in the presence of Al^{3+} compared with the tested competing ions; however, the absence of quantitative selectivity coefficients and realistic mixed-ion experiments limits conclusions regarding performance in complex real samples.

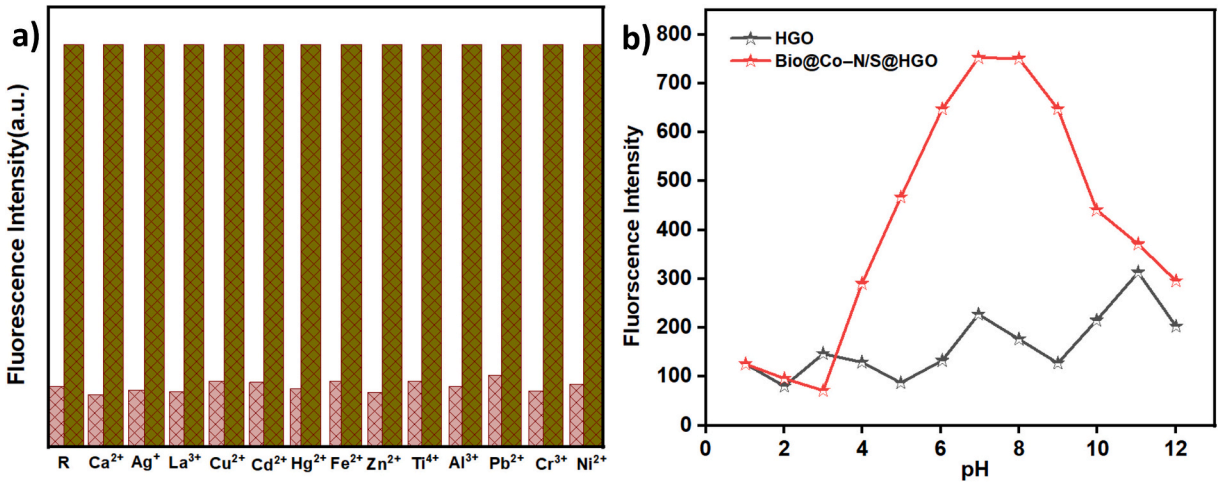


Fig. 10. (a) Interference studies, (b) effect of pH.

CRediT authorship contribution statement

M. Kalaiyarasi: Writing – original draft, Methodology, Conceptualization. **Pugalenthi Ramesh:** Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Sarah A. Alshehri:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **K. Gilbert Ross Rex:** Writing – review & editing, Formal analysis, Data curation. **K. Gopi:** Validation, Formal analysis, Data curation. **V. Manikandan:** Writing – review & editing, Project administration, Formal analysis, Data curation. **Ganesh Koyyad:** Writing – review & editing, Validation, Resources. **Tamanna Pradhan:** Writing – review & editing, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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